# Benchmarking of the AES symmetric encryption algorithm on MicroBlaze softcore processor

Jack Keefer, Elias Cortes, Adam Tam, Michelle Lau
*Electrical and Computer Engineering Department*
*California State Polytechnic University, Pomona*
{jtkeefer, ecortes, adtam, mllau}@cpp.edu

***Abstract*—The Advanced Encryption Standard (AES) is a widely used symmetric encryption algorithm that provides secure data encryption. This paper presents a benchmarking study of the AES algorithm implemented on a MicroBlaze softcore processor, hosted on a Nexys XC7A100T FPGA. The performance of the AES implementation is evaluated in terms of throughput, latency, and energy expenditure. By maintaining a strictly constant physical hardware footprint, this study isolates the algorithmic penalty of expanded key schedules and additional encryption rounds. The results demonstrate the critical trade-offs between higher security tiers and increased processing delays, showcasing the necessity of identifying the correct balance for resource-constrained edge and IoT applications.**

***Index Terms*—Advanced Encryption Standard (AES), Benchmarking, Cryptography, Field Programmable Gate Array (FPGA),**

## I. Introduction

The proliferation of edge computing nodes and IoT devices has fundamentally shifted the landscape of digital security. As sensitive data is increasingly processed at the network edge, lightweight encryption solutions are required to ensure data integrity. The Advanced Encryption Standard (AES) is a widely used symmetric encryption algorithm that provides secure data encryption at three security levels: AES-128, AES-192, and AES-256. However, there is an inherent trade off between higher security and increased resource utilization, a particularly critical consideration for resource-constrained embedded systems.

While AES-128 is currently considered secure against classical attacks, the advent of quantum computing poses a significant threat to its security. AES-256, with its larger key size and increased number of rounds, offers enhanced security margins against potential quantum attacks. However, this increased security comes at the cost of higher computational complexity, which can lead to increased latency and energy consumption.

This paper presents a benchmarking study of the AES algorithm implemented on a MicroBlaze softcore processor, hosted on a Nexys XC7A100T FPGA. To accurately quantify the costs of scaling security, the AES-128, AES-192 and AES-256 algorithms were compiled and executed bare-metal on the MicroBlaze core. The performance of the AES implementation is evaluated in terms of throughput, latency, and energy expenditure. These results help inform embedded system designers of the trade-offs associated with selecting higher security tiers, enabling data-driven decisions for hardware provisioning in resource-constrained edge and IoT environments where quantum threats may not be an immediate concern.

## II. Background and Related Work

While quantum computing is not yet a viable attack vector against edge devices, the potential for future quantum attacks requires a proactive approach to security. A common concern is harvest-now-decrypt-later attacks, where adversaries collect encrypted data today with the intention of decrypting it in the future once quantum capabilities are available. This however doesn't mean that AES-256 is the default choice for all applications. The increased computational overhead of AES-256 can lead to significant performance degradation. Prior research has shown that AES-128 is suitable for current edge applications, while AES-256 may be reserved for scenarios where long-term security is paramount and performance constraints are less critical [1], [2]. Additionally concerns over future quantum attacks may be unnecessary for many applications as research has shown that AES-128 is expected to remain secure against quantum attacks for the foreseeable future, with estimates suggesting that it would require an impractically large quantum computer to break AES-128 within a reasonable timeframe due to the difficulty of implementing Grover's algorithm at scale [3].

## III. Methodology

### A. Hardware Setup

The experiment was conducted using a Digilent Nexys A7-100t Training Board, featuring the Xilinx Artix-7 XC7A100T FPGA fabric [4]. This hardware was selected due to availibility as a required component of other classes in the Electrical and Computer Engineering program. The specifications of the development board are summarized in Table I.

TABLE I
Hardware Specifications of the Training Board

| Parameter | Nexys A7-100T |
|---|---|
| FPGA part | XC7A100T-1CSG324C |
| Logic Slices | 15,850 |
| Block RAM (Kbits) | 4,860 |
| DDR2 Memory (MiB) | 128 |
| Clock Tiles (with PLL) | 6 |
| DSP Slices | 240 |

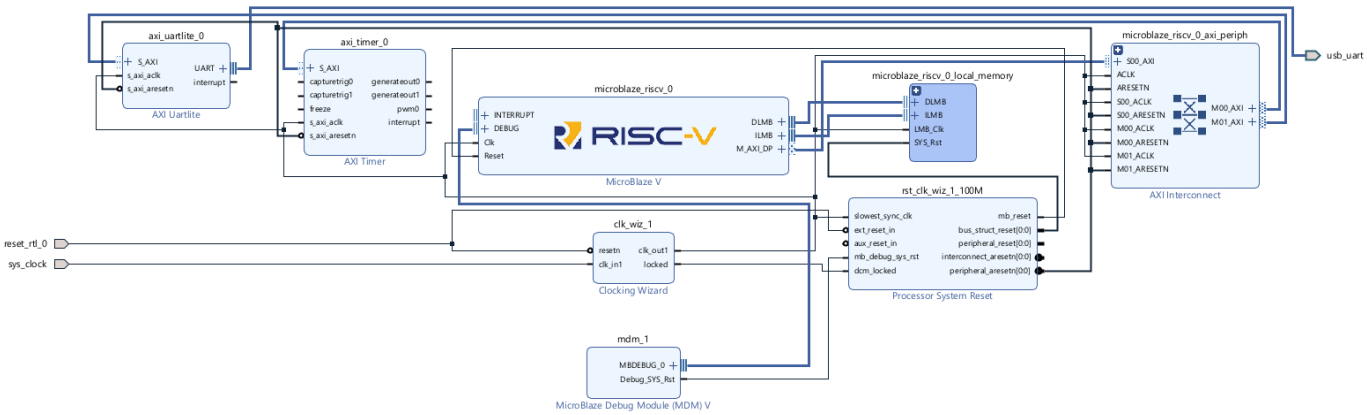


Fig. 1. Block diagram of the AES implementation on the MicroBlaze softcore processor.

### B. Software Implementation

The AES-128, AES-192, and AES-256 algorithms were compiled and executed bare-metal on the MicroBlaze core as shown in Figure 1. Cycle counts were captured using internal hardware timers. Power estimation was performed using the Vivado Power Analyzer post-implementation to derive dynamic and static power dissipation figures shown in Table II.

TABLE II
BASELINE HARDWARE UTILIZATION AND POWER CHARACTERISTICS OF THE MICROBLAZE SOFTCORE IMPLEMENTATION

| Metric | Value |
|---|---|
| *Hardware Utilization* | |
| LUTs | 2,742 |
| Flip-Flops (FFs) | 2,161 |
| BRAM | 16 |
| DSPs | 4 |
| *Power Consumption* | |
| Total Power (W) | 0.229 |
| Dynamic Power (W) | 0.131 |
| Static Power (W) | 0.098 |

## IV. RESULTS AND ANALYSIS

As Microblaze is a softcore processor, the hardware utilization metrics and power consumption remain constant across all AES key sizes. Energy data was collected using the Vivado Power Analyzer and is based on the post-implementation design, which includes the MicroBlaze core and its associated peripherals. These baseline metrics are summarized in Table II.

### A. Resource Utilization and Memory Footprint

Post-implementation reports from Vivado confirm a static hardware footprint across all three security tiers. The MicroBlaze core and its associated peripherals utilized the static logic and memory resources detailed in Table II. While the physical footprint remained fixed the memory footprint—represented by the code size—increased slightly from AES-128 to AES-256 as shown in Table III. This increase is directly attributable to the larger arrays required to store the expanded round keys in the processor's memory.

TABLE III
PERFORMANCE AND ENERGY METRICS BY AES KEY SIZE

| Metric | AES-128 | AES-192 | AES-256 |
|---|---|---|---|
| *Performance* | | | |
| Cycles | 40,246 | 48,384 | 55,767 |
| Execution Time (µs) | 402.46 | 483.84 | 557.67 |
| Throughput (kbps) | 318.00 | 264.55 | 229.53 |
| Cycles per Byte (CpB) | 2,515.38 | 3,024.00 | 3,485.44 |
| Code Size (bytes) | 18,037 | 18,065 | 18,193 |
| *Energy Efficiency* | | | |
| Energy per Enc. (µJ) | 52.72 | 63.38 | 73.05 |
| Energy per Bit (nJ/bit) | 411.88 | 495.16 | 570.70 |
| *Hardware Trade-offs* | | | |
| Throughput/Area (kbps/LUT) | 0.116 | 0.096 | 0.084 |
| Area-Time Product (LUT-µs) | 1,103,545 | 1,326,689 | 1,529,131 |

### B. Execution Latency and Throughput

The most significant impact of scaling the AES key size was observed in the execution latency. Transitioning from the baseline AES-128 algorithm to AES-256 resulted in a notable increase in execution time and required clock cycles, as detailed in Table III and shown in Figure 2. This proportional increase aligns with the increased algorithmic complexity of AES; the transition from 10 to 14 encryption rounds demands more processor instructions per 16-byte plaintext block. Consequently, the overall throughput showed an inverse relationship with key size. To normalize these performance metrics against the block size, the computational cost was calculated in Cycles per Byte (CpB) with the following formula and summarized in Table III and shown in Figure 3:

$$CpB = \frac{\text{Total Cycles}}{\text{Block Size (Bytes)}} \tag{1}$$

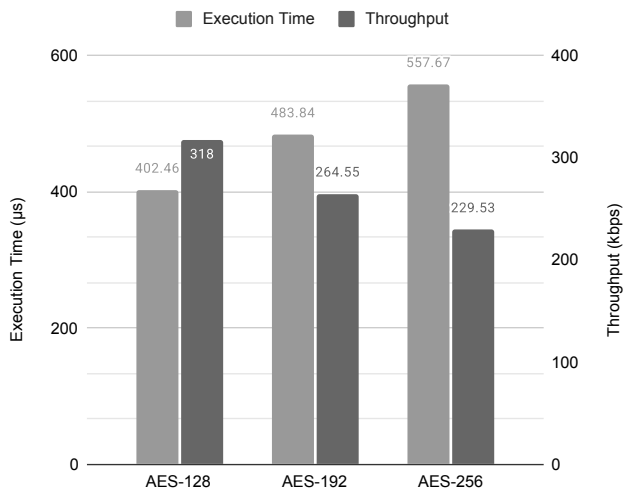


Fig. 2. Throughput and execution time for AES-128, AES-192, and AES-256 implementations on the MicroBlaze softcore processor.

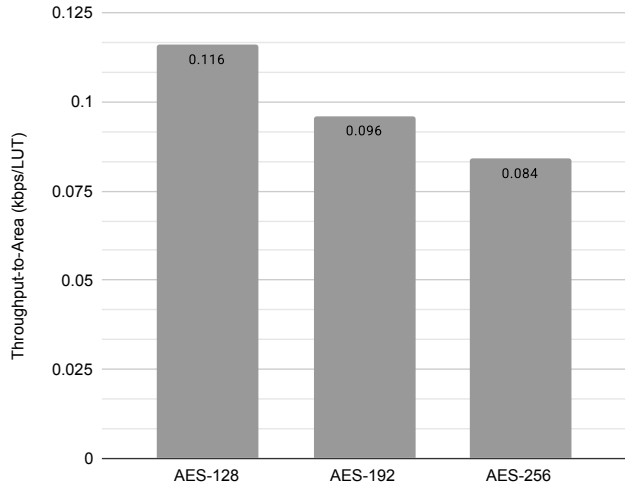


Fig. 4. Throughput-to-Area ratio (kbps/LUT) for AES-128, AES-192, and AES-256 implementations on the MicroBlaze softcore processor.

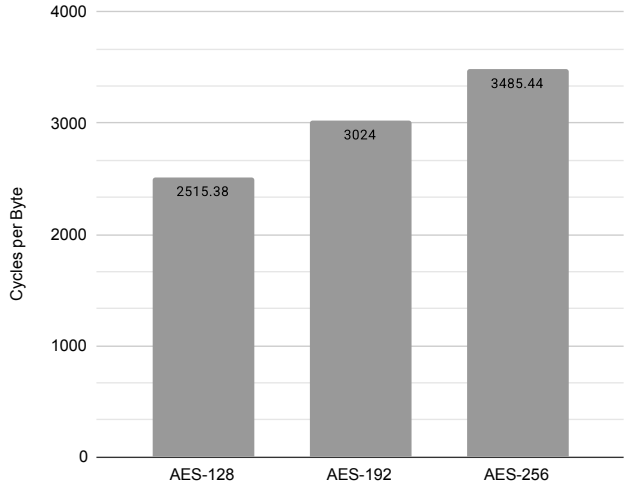


Fig. 3. Cycles per Byte (CpB) for AES-128, AES-192, and AES-256 implementations on the MicroBlaze softcore processor.

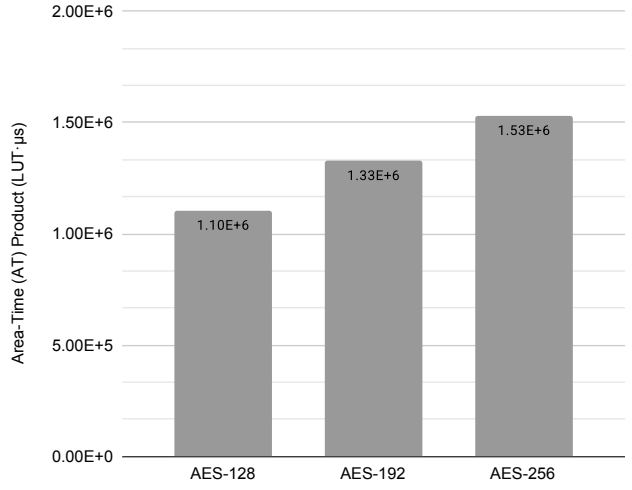


Fig. 5. Area-Time (AT) Product for AES-128, AES-192, and AES-256 implementations on the MicroBlaze softcore processor.

### C. Architectural Efficiency

To evaluate how effectively the static logic resources were utilized under varying cryptographic workloads, the Throughput-to-Area ratio was derived and summarized in Table III and shown in Figure 4. Furthermore, the Area-Time (AT) Product was calculated to quantify the fundamental hardware penalty of the implementations with the following formula and is also summarized in Table III and shown in Figure 5:

$$AT = \text{LUTs} \times \text{Execution Time } (\mu)\text{s} \tag{2}$$

This mathematical increase confirms that while the physical area did not change, the "time-area" cost of occupying the processor for longer durations imposes a heavier system penalty for stricter security.

### D. Energy and Power Profiling

Power analysis was conducted using the Vivado Power Analyzer, which reported a constant total power dissipation (comprising dynamic and static power) across all variants due to the static nature of the underlying processor, as seen in Table II. However, because execution time varied, the actual dynamic energy consumed per encryption event scaled significantly with the key size. For system-level power budgeting, this was converted to Energy per Bit ($E_{\text{bit}}$) using the following formula and summarized in Table III and shown in Figure 6:

$$E_{\text{bit(dynamic)}} = \frac{\text{Energy per Encryption}}{128 \text{ bits}} \tag{3}$$

This demonstrated an increase in energy cost per bit as the key sized increased. This is a result of the longer execution time required for AES-256, which results in more energy being consumed while the processor is active, despite the static power remaining unchanged. This demonstrates the significant energy trade-offs associated with higher security levels on a fixed hardware platform.

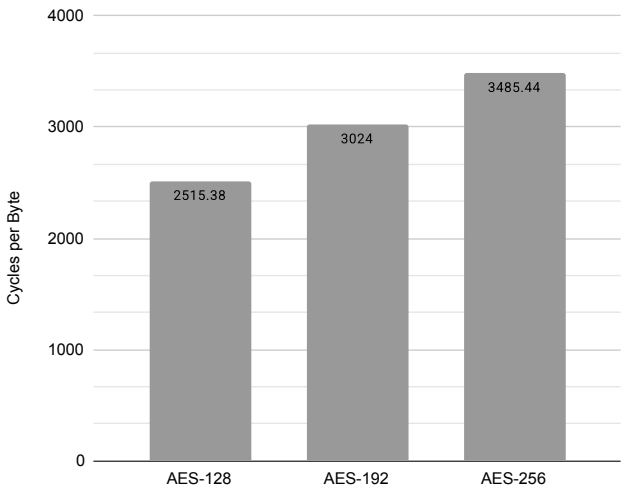


Fig. 6. Energy per Bit (nJ/bit) for AES-128, AES-192, and AES-256 implementations on the MicroBlaze softcore processor.

## V. Conclusion

This study demonstrates the architectural and energy trade-offs of scaling the Advanced Encryption Standard (AES) across 128-bit, 192-bit, and 256-bit key sizes on a 32-bit MicroBlaze soft processor. By executing the cryptographic algorithms entirely in software on the Xilinx Artix-7 FPGA fabric, the physical hardware utilization remained strictly constant. This static hardware baseline successfully isolated the algorithmic penalty of the expanded AES key schedules and additional encryption rounds.

The benchmarking results demonstrate a deterministic, inverse relationship between cryptographic security and system efficiency. Upgrading from the base AES key size of 128 bits to AES-256 increased execution latency, causing a proportional throughput degradation. Furthermore, the Area-Time (AT) Product scaled significantly, proving that higher security tiers impose a heavier processing delay penalty on the soft core. Energy profiling revealed that while the system's static power draw remained stable, the dynamic algorithmic cost per bit increased linearly with key size. The transition to AES-256 resulted in a substantial dynamic energy penalty per encrypted bit compared to the AES-128 baseline. With limited evidence to support 128-bit AES being vulnerable to quantum attacks in the near future, these results suggest that AES-128 may be the optimal choice for many resource-constrained edge and IoT applications where performance and energy efficiency are critical. However, for scenarios where long-term security is paramount and performance constraints are less critical, AES-256 may be justified despite its increased overhead.

## References

[1] E. Barker and A. Roginsky, "Transitioning the use of cryptographic algorithms and key lengths," 2019.

[2] J. P. Mattsson, "Post-quantum cryptography in mobile networks," Ericsson Blog, jun 2017. [Online]. Available: https://www.ericsson.com/en/blog/2017/6/post-quantum-cryptography-in-mobile-networks

[3] S. Shahmir, G. Murtaza, Ala-Al-Fuqaha, S. Al-Kuwari, and T. Abbas, "Resource constraint estimation of in-quantum implementated mini-aes," *arXiv*, 2025.

[4] P. by Digilent. Nexys a7 amd artixtm 7 fpga trainer board: Recommended for ece curriculum. [Online]. Available: https://digilent.com/shop/nexys-a7-amd-artix-7-fpga-trainer-board-recommended-for-ece-curriculum/